

Experiment 4

AIM: Use npm / yarn / pip to install and manage packages

Step 1: Check Installation

```
C:\Users\kisha>node -v
```

```
v22.13.1
```

```
C:\Users\kisha>npm -v
```

```
10.9.2
```

Step 2: Create New npm Project

```
C:\Users\kisha>mkdir npm-lab
```

```
C:\Users\kisha>cd npm-lab
```

```
C:\Users\kisha\npm-lab>npm init -y
```

Wrote to C:\Users\kisha\npm-lab\package.json:

```
{  
  "name": "npm-lab",  
  "version": "1.0.0",  
  "main": "index.js",  
  "scripts": {  
    "test": "echo \"Error: no test specified\" && exit 1"  
  },  
  "keywords": [],  
  "author": "",  
  "license": "ISC",  
  "description": ""  
}
```

Step 3: Install Required Packages

```
C:\Users\kisha\npm-lab>npm install express  
added 65 packages, and audited 66 packages in 10s  
22 packages are looking for funding  
run `npm fund` for details  
found 0 vulnerabilities
```

```
C:\Users\kisha\npm-lab>npm install axios  
added 11 packages, and audited 77 packages in 5s  
24 packages are looking for funding  
run `npm fund` for details  
found 0 vulnerabilities
```

```
C:\Users\kisha\npm-lab>npm install nodemon --save-dev  
added 27 packages, and audited 104 packages in 4s  
28 packages are looking for funding  
run `npm fund` for details  
found 0 vulnerabilities
```

```
C:\Users\kisha\npm-lab>npm list  
npm-lab@1.0.0 C:\Users\kisha\npm-lab  
+-- axios@1.13.5  
+-- express@5.2.1  
`-- nodemon@3.1.11
```

Step 4: Create Web Server

Create a file named index.js and Add below content :

```
const express = require("express");
const axios = require("axios");
const app = express();

app.get("/", (req, res) => {
  res.send("Hello Open Source Lab");
});

// JSON Route
app.get("/api/info", (req, res) => {
  res.json({
    name: "NPM Lab",
    subject: "Open Source Technology",
    semester: "BE 6th Sem" });
});

// Axios Demo Route
app.get("/api/user", async (req, res) => {
  const response = await axios.get("https://jsonplaceholder.typicode.com/users/1");
  res.json(response.data);
});

app.listen(3000, () => {
  console.log("Server running at http://localhost:3000");
});
```

Step 5: Run Server (Normal Way)

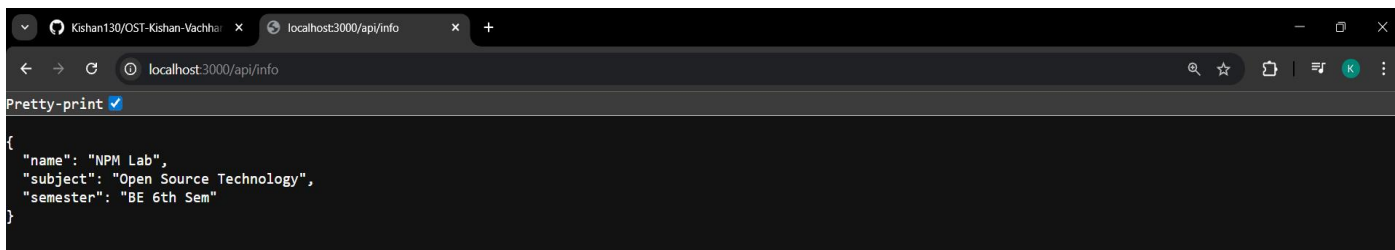
C:\Users\kisha\npm-lab>node index.js

Server running at <http://localhost:3000>

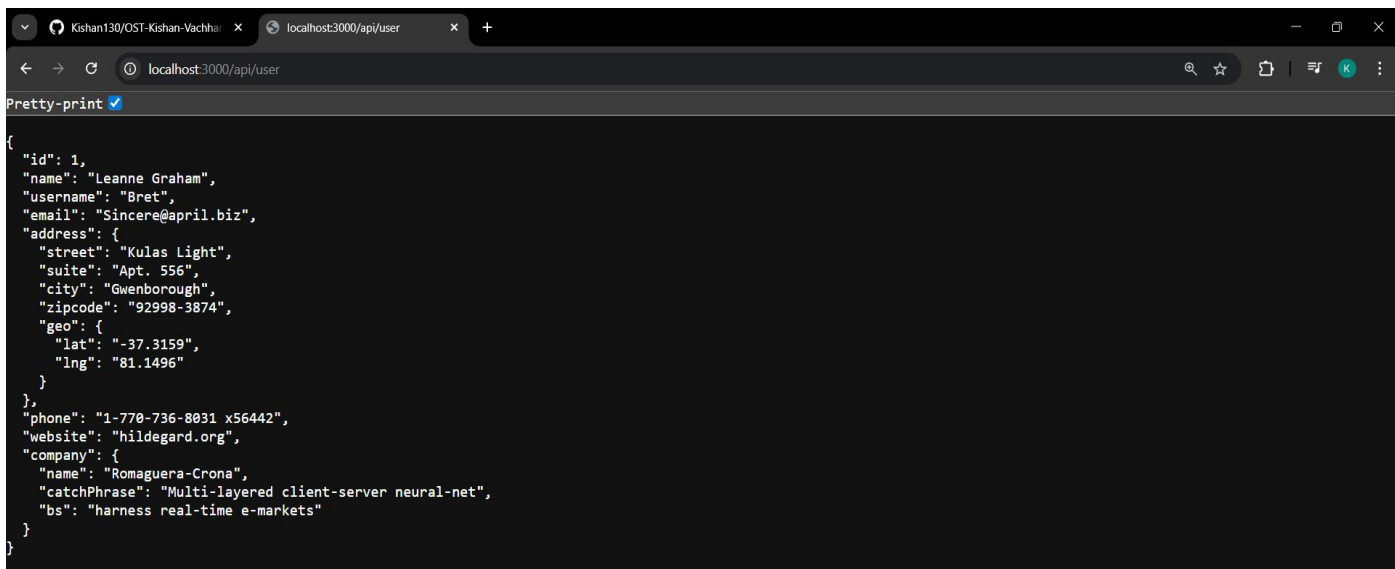
<http://localhost:3000>



<http://localhost:3000/api/info>



<http://localhost:3000/api/user>



Step 6: Add npm Scripts

Open package.json and modify scripts section:

```
"scripts": {  
  "start": "node index.js",  
  "dev": "nodemon index.js"  
}
```

Step 7: Run Using npm Script

1. Normal mode:

```
C:\Users\kisha\npm-lab>npm run start
```

```
> npm-lab@1.0.0 start  
> node index.js
```

```
Server running at http://localhost:3000  
Terminate batch job (Y/N)? y
```

2. Development mode (nodemon):

Now modify index.js text and save → Server automatically restarts (nodemon demo).

```
C:\Users\kisha\npm-lab>npm run dev
```

```
> npm-lab@1.0.0 dev  
> nodemon index.js
```

```
[nodemon] 3.1.11
```

```
[nodemon] to restart at any time, enter `rs`
```

```
[nodemon] watching path(s): *.*
```

```
[nodemon] watching extensions: js,mjs,cjs,json
```

```
[nodemon] starting `node index.js`
```

```
Server running at http://localhost:3000
```

[nodemon] restarting due to changes...

[nodemon] starting `node index.js`

Server running at http://localhost:3000

[nodemon] restarting due to changes...

[nodemon] starting `node index.js`

Server running at http://localhost:3000

Terminate batch job (Y/N)? y

Step 8: Delete node_modules and Reinstall

Delete manually

Now reinstall:

C:\Users\kisha\npm-lab>npm install

added 103 packages, and audited 104 packages in 5s

28 packages are looking for funding

run `npm fund` for details

found 0 vulnerabilities

Step 9: Add .gitignore File

Create file: .gitignore

Add:

node_modules/



Step 10: Upload Project to Git (Without node_modules)

```
C:\Users\kisha\npm-lab>git init
```

Reinitialized existing Git repository in C:/Users/kisha/npm-lab/.git/

```
C:\Users\kisha\npm-lab>git add .
```

```
C:\Users\kisha\npm-lab>git commit -m "Practical 4 Commit"
```

On branch main

nothing to commit, working tree clean

```
C:\Users\kisha\npm-lab>git remote add origin https://github.com/Kishan130/OST-Kishan-Vachhani
```

error: remote origin already exists.

```
C:\Users\kisha\npm-lab>git branch -M main
```

```
C:\Users\kisha\npm-lab>git push -u origin main
```

Enumerating objects: 9, done.

Counting objects: 100% (9/9), done.

Delta compression using up to 6 threads

Compressing objects: 100% (7/7), done.

Writing objects: 100% (8/8), 12.56 KiB | 1.57 MiB/s, done.

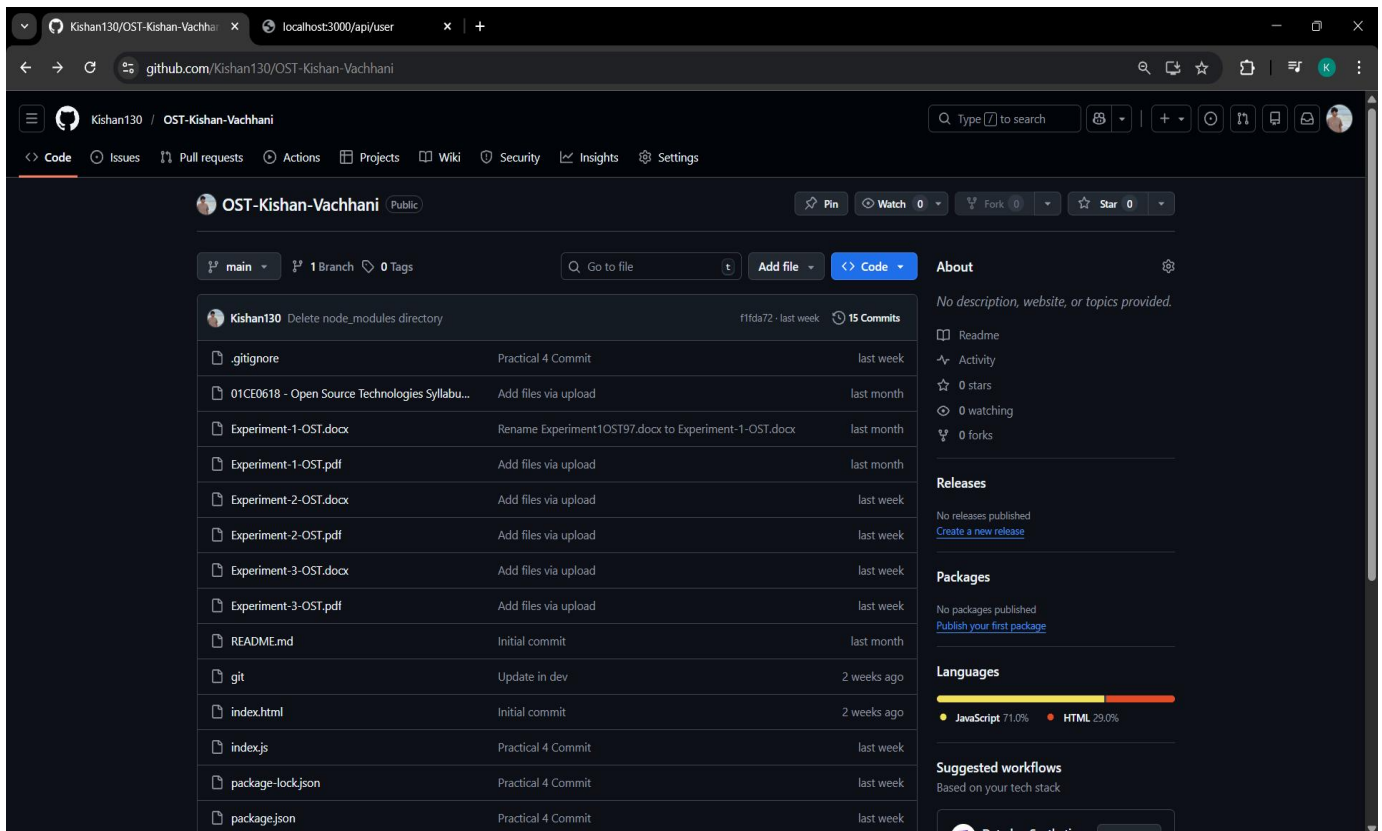
Total 8 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)

remote: Resolving deltas: 100% (3/3), completed with 1 local object.

To <https://github.com/Kishan130/OST-Kishan-Vachhani.git>

e55b01e..ba37fdd main -> main

branch 'main' set up to track 'origin/main'.



```
C:\Users\kisha\npm-lab>git push -u origin main
Enumerating objects: 1195, done.
Counting objects: 100% (1195/1195), done.
Delta compression using up to 12 threads
Compressing objects: 100% (1119/1119), done.
Writing objects: 100% (1194/1194), 1.40 MiB | 687.00 KiB/s, done.
Total 1194 (delta 174), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (174/174), completed with 1 local object.
To https://github.com/Kishan130/OST-Kishan-Vachhani
7c38190..4f2188c main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\kisha\npm-lab>
```